



REVIEW

Root and root-derived biopore interactions in soils: A review

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Abstract

In recent years, the investigation of root–biopore interactions has been reported in many studies, particularly with the application of X-ray computed tomography (CT). To clarify our understanding of root–biopore interactions, we reviewed the effects of plant roots on soil structure and the biopore effects on plant growth and uptake of nutrients and water. Plant roots can penetrate the soil matrix and form biopores after decay. The biopores are cylindrical, continuous, and stable and facilitate the transport of air and water. The soil around biopores also affects plant growth by its rich nutrients; hence, the concept of a biopore sheath is proposed to indicate the soil affected by the roots and resulting biopores. The behavior of roots encountering biopores is impacted by soil strength, oxygen concentration, biopore characteristics, and crop species. The biopores provide many channels with the least mechanical resistance for crop roots to enter the subsoil, improving the absorption of water and nutrients by crops. However, the effect of biopores on crop growth depends on biopore size. The smaller biopores (<2 mm) promoted plant growth, but the larger biopores did not (≥ 2 mm), which was mainly related to the degree of contact between root and soil in biopores. However, quantification of root–biopore wall contact is difficult due to the low resolution of CT images, which limits the understanding of the effects of biopore characteristics on plant growth and nutrient uptake. Therefore, determining the contact area between roots and soil in the biopores and revealing the contribution of biopore characteristics to water and nutrient uptake are important to give full play to the functions of biopores in sustainable agriculture.

KEYWORDS

biopore, biopore sheath, plant roots, soil structure, X-ray computed tomography

1 | INTRODUCTION

In recent decades, soil compaction has become a global problem due to increasing agricultural mechanization, leading to an increase in soil bulk density or a decrease in porosity, which significantly limits the growth of roots, reducing the yield of crops (Correa et al., 2019). A meta-analysis by Obour and Ugarte (2021) reported that 6%–34% grain yields were decreased by soil compaction. The plant roots have difficulty in reaching the compacted subsoil due to high mechanical impedance (Bengough et al., 2011), which limits the root utilization of nutrients and water in the subsoil. Once the topsoil is depleted of nutrients and water, the growth of crops will be severely hindered. However, it has been reported that cover crops with deep and strong

root systems are able to penetrate compacted subsoil and create many biopores when the roots are decomposed; this process is termed “bio-tillage” (Zhang & Peng, 2021). Pulido-Moncada et al. (2022) found that cover crops such as lucerne could create many vertical and well-connected biopores in compacted subsoil. Zhang et al. (2022) reported that bio-tillage using cover crops improved air permeability and root growth and increased yield of succeeding maize in compacted soil. This indicates that biopores formed by bio-tillage play a significant role in promoting plant growth in compacted soil. Thus, the utilization of plant roots to improve soil pore structure receives much attention globally (Lucas, 2022).

The radial pressure caused by roots penetrating the soil may compress the surrounding soil (Dexter, 1987), leading to a decrease in

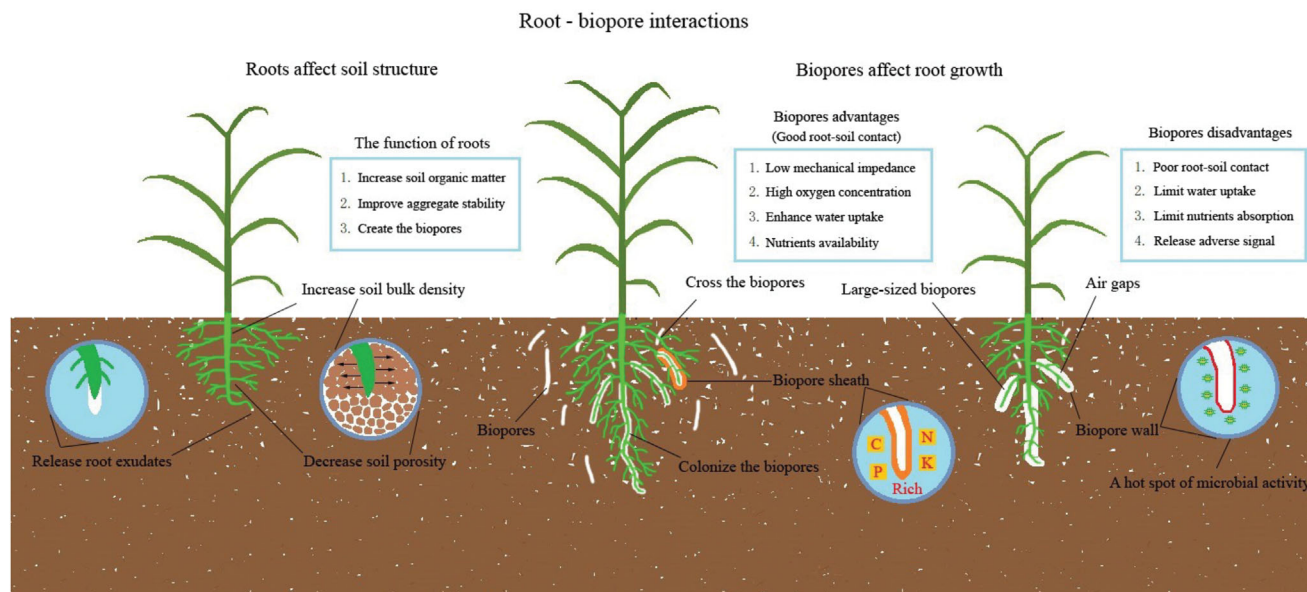


FIGURE 1 The effects of plant roots on soil structure. The advantages and disadvantages of biopores, and behaviors of roots encountering the biopores.

porosity in the rhizosphere (Bruand et al., 1996). Plant roots can improve soil structural stability in the rhizosphere by gelling soil particles with root exudates (Traoré et al., 2000; Naveed et al., 2017). An increase in soil aggregation was also observed near the roots of the plant (Angers & Caron, 1998; Dormaar & Foster, 1991). In addition, many researchers have reported that crop roots can create channels in the process of growth, which then decompose to form biopores (Cresswell & Kirkegaard, 1995; Han, Kautz, Perkons, Lüsebrink, et al., 2015). Angers and Caron (1998) suggested that one of the most critical changes in soil structure caused by plant roots was the formation of connected biopores.

There are many natural biopores existing in the soil, which have a wide range of functions (Kautz, 2015). Biopores can facilitate air and water transport in soils (Dziejowski et al., 1997; Naveed et al., 2016). Many studies have revealed that roots grow preferentially toward biopores in unfavorable soil conditions (Atkinson et al., 2020; Colombi et al., 2017; Pfeifer et al., 2014; Stirzaker et al., 1996). When the soil is hard, the presence of biopores can provide some of the least resistance pathways for roots to reach the subsoil (Athmann et al., 2013; Colombi et al., 2017; White & Kirkegaard, 2010). Thus, these biopores are beneficial for roots to absorb nutrients and water in the subsoil (Han et al., 2017; Kautz, 2015; McKenzie et al., 2009). Moreover, the biopore sheath is usually rich in nutrients due to the influence of root exudates and decay of roots (Pierret et al., 1999). For example, the biopore wall contains a higher concentration of organic C and total N than those in the bulk soil, resulting in the biopore wall being a hot spot for microbial activity (Tiunov & Scheu, 1999). Plant roots prefer to grow in the soil near the biopores compared with the soil far away from the biopores (Petzoldt et al., 2020).

Plant roots are capable of improving soil pore structure, and root-derived biopores and biopore sheaths might have a significant effect on

plant root growth (Angers & Caron, 1998; Kautz, 2015; Petzoldt et al., 2020). However, root-biopore interactions have not been specifically reviewed. Although there are some studies investigating the effects of biopores on plant growth, a comprehensive summary of the root-biopore interactions from these studies is lacking, which is valuable for improving our understanding of biopore effects on crop growth. In addition to the biopores, the biopore sheath is equally important and should be considered. Thus, this review summarizes the current knowledge about the effects of plant roots on soil structure, and the responses of roots to biopores.

2 | THE FUNCTION OF PLANT ROOTS

2.1 | Plant roots improve soil structure

As shown in Figure 1, the presence of complete root system architecture in soil can considerably improve soil structure (Reubens et al., 2007). When roots grow in the soil, the elongation of root tips in the soil matrix results in compression of the soil and a decrease in soil porosity around the roots (Dorioz et al., 1993; Koebernick et al., 2017; Phalempin et al., 2021). Relative to the bulk soil, the soil bulk density near the roots is generally higher (Young, 1998). However, Helliwell et al. (2019) observed an increase in porosity at the root-soil interface due to soil and root shrinkage and thigmotropic responses in roots, despite the penetration of roots resulting in the tight arrangement of soil particles. Koebernick et al. (2019) also reported an increase in soil porosity near the root surface, which was induced by the incomplete packing of soil particles on the root surface. Materechera, Dexter, et al. (1992) found that water uptake by plant roots can also change the soil structure by affecting the soil water content. Colombi et al. (2021)

reported that root water uptake could result in the formation of soil cracks in the vicinity of roots. This process will increase the connectivity of pores, especially when biopores are formed (Galdos et al., 2019; Pulido-Moncada et al., 2020). Moreover, Zhang et al. (2021) found that microbial activity resulting from root exudates increased the proportion of small pores with diameters $<60\text{ }\mu\text{m}$, which improved soil hydraulic conductivity. Similarly, Feeney et al. (2006) also observed that the presence of microbes near the roots might increase soil porosity and change pore geometry. However, Kroener et al. (2014) found that saturated hydraulic conductivity of soil decreased with increasing concentrations of mucilage exuded by roots, which was attributable to a higher viscosity of mucilage.

Plant roots can also improve soil aggregation by root exudates and stimulated microbial activities (Golchin et al., 1994). Czarnes et al. (2000) found that root exudates could stabilize the soil structure by improving the strength of interparticle binding. Moreover, the penetration of plant roots can create channels, and mucilages from roots enhance the strength of the channels, which establish a favorable condition for the formation of stable biopores (Ghestem et al., 2011). When plant roots die and begin to decompose, root-derived particulate organic matter is wrapped by clay particles and microbial binding agents, promoting the formation of soil organic matter and aggregates (Cotrufo et al., 2013; Six et al., 2004), and the remaining channels are biopores (Yunusa & Newton, 2003).

To assess the effects of plant roots on soil structure, we summarized the responses of aggregate stability, air permeability, saturated hydraulic conductivity, and macroporosity to crop roots from 20 studies (Table 1). The result suggests that aggregate stability was improved by plant roots in eight of 11 studies, while three of 11 studies did not. Cover crop roots increased air permeability in three of four studies and were not affected in one study. Saturated hydraulic conductivity was enhanced by roots in five of eight studies but not in three of eight studies. Moreover, the roots increased macroporosity in six of seven studies but not in only one study. In summary, the presence of plant roots generally improves air permeability, saturated hydraulic conductivity, macroporosity, and aggregate stability.

2.2 | The characteristics of biopores formed by roots

The characteristics of crop roots have an important effect on the formation of biopores in arable fields (Materechera, Alston et al., 1992). The large-sized biopores increase when deep root plants have a larger root diameter (Han et al., 2017), and the density of small-sized biopores increases when the plant, such as tall fescue, has a smaller root diameter (Huang & Gao, 2000). Yunusa and Newton (2003) reported that the diameter of biopores was between $30\text{ }\mu\text{m}$ and 5 mm , which facilitated air and water transport. Root-derived biopores may vary in size and shape due to a variety of architectures of the root system (Bodner et al., 2021). Zhang et al. (2018) observed that the shape of biopores created by roots was usually cylindrical and continuous, while non-biopores were irregular and discontinuous. As a result, biopores are

much more effective in transporting water and air than non-biopores (Luo et al., 2008; Zhang et al., 2019). Root-derived biopores are more stable than non-biopores and can persist for decades in the subsoil (Hagedorn & Bundt, 2002). However, the shape of non-biopores, such as cracks, varies dynamically with the soil moisture content. Normally, cracks significantly reduce or completely close under continuous flooding conditions (Novak et al., 2000; Zhang et al., 2014). Unlike cracks, which are strongly affected by drying and wetting cycles, biopore structures are rarely influenced by these cycles and are much more stable (Cheik et al., 2021). Schäffer et al. (2008) also found that biopores were more stable than interaggregate pores because they were less affected by soil compaction and could continue to be effective as other pores collapsed.

Although earthworm burrows are also cylindrical and continuous in shape, their tortuosity is quite different from that of root-derived biopores (Pagenkemper et al., 2013). Some studies have indicated that tortuosity is much greater for earthworm burrows than for root-derived biopores, which can be used as an indicator to distinguish between root-derived biopores and earthworm burrows (Bastardie et al., 2002; Langmaack et al., 2002; Zhang et al., 2018). Furthermore, Bottinelli et al. (2017) found that earthworm burrows might become shorter and discontinuous due to the refilling of casts in the burrows. In summary, the characteristics of root-derived biopores are distinct from those of earthworm burrows and non-biopores, which may influence their performance in air and water conductivity and crop growth.

2.3 | Definition of the biopore sheath

The rhizosphere is defined as the soil near living roots and is affected by the activity of roots (Hartmann et al., 2008; Hiltner, 1904; Hinsinger et al., 2009). When plant roots die, the concepts of the rhizosphere may not be applicable due to the cessation of root activity. Thus, a new concept needs to be developed. We propose the “biopore sheath,” which means the soil is affected by roots and resulting biopores. These effects include not only the enrichment of nutrients and microorganisms in the biopore sheath soil (Athmann et al., 2017; Tiunov & Scheu, 1999), but also changes in its structure during the formation of biopores. The biopore sheath is no longer influenced by root exudates after root death, which may reduce their structural stability (Czarnes et al., 2000) and water holding capacity (York et al., 2016). However, the biopore sheath is greatly influenced by preferential flow after root decay, which rarely occurs in the rhizosphere. Luo et al. (2008) observed that the solute would be transported laterally from the biopores to the biopore sheath when preferential flow formed. Morales et al. (2010) reported that preferential flow channels (biopore sheaths) were rich in organic matter mainly from dissolved organic matter, decaying roots, and microbial biomass carbon. Preferential flow channels act as hot spots for microbial activity (Bundt et al., 2001), with exudates from microorganisms enhancing the cohesion between soil particles, thereby contributing to the stabilization of the biopore sheath (Vu et al., 2022). Although the biopore sheath is derived from the rhizosphere, its properties are quite different from those of the rhizosphere,

TABLE 1 Effects of plant roots on soil structure.

Location	Soil texture	Plant species	Duration (year)	Depth (cm)	AS (%)	K_a (μm^2)	K_s (mm h^{-1})	MP (%)	Reference
Denmark	Sandy loam	Chicory Lucerne Tall fescue	2	0–10		ns ns ns		ns I ns	Pulido-Moncada et al. (2022)
China	Clay loam	Alfalfa Oilseed rape Radish + vetch	2	30–50		I ns ns	I ns I		Zhang et al. (2022)
Brazil	Sandy clay	Sorghum-grass Sunn hemp millet	3	5–10	I ns I			I I I	Garcia et al. (2013)
Germany	Silt loam	Oil radish Saia oat Spring vetch Winter rye	1	0–10	I ns ns I				Grunwald et al. (2022)
USA	Silt loam	Winter lentil Spring lentil spring pea Winter triticale Spring triticale Winter wheat	5	0–7.5	ns I I I I				Blanco-Canqui et al. (2013)
USA	Silt loam	Winter wheat	2	0–60	I				Adeli et al. (2019)
Brazil	Clay	Vetch Lupin Radish Oat Wheat	19	0–60	I I I I I			ns ns ns ns ns	Calegari et al. (2013)
USA	Silt loam	Cereal rye + Hairy vetch + Winter pea	1	0–10			ns	I	Cercioglu et al. (2018)
USA	Silt loam	Cereal rye	4	0–40			ns	I	Haruna et al. (2018)
USA	Clay loam	Juan triticale + Merced rye + Common vetch	15	0–15	I				Mitchell et al. (2017)
USA	Silty clay loam	Pea + turnip	1	0–10	ns				Mukherjee and Lal (2015)
USA	Silt loam	Winter wheat	37	0–15	ns		ns		Nouri et al. (2018)
USA	Silt loam	Hairy vetch winter wheat	34	0–15	ns ns		I I		Nouri et al. (2019)
Argentina	Silt loam	Oats Barley Ryegrass Rescue grass Vetch Oats + vetch Forage radish Rapeseed	6	0–5	I I I I I I I I			I I I ns I I ns I	Restovich et al. (2019)
Brazil		Millet Sunn hemp	1	20–30				I ns	Teixeira et al. (2016)
Spain	Loam	Barley Vetch	10	0–5	I ns		I ns		García-González et al. (2018)
USA	Silt loam	Rye	2	5–10			I		Liesch et al. (2011)
Canada	Clay loam	Winter wheat	5	0–20			I		Drury et al. (2014)
Denmark	Sandy loam	Fodder radish	5	4–8		I			Abdollahi et al. (2014)
USA	Silt loam	Cereal rye	13	0–7		I			Steele et al. (2012)

Note: The roots reduce and increase soil parameters are represented by "R" and "I," respectively.

Abbreviations: AS, aggregate stability; K_a , air permeability; K_s , saturated hydraulic conductivity; MP, macroporosity; ns, not significant.

which should be investigated independently from the rhizosphere. The rhizosphere is usually 0–2 mm from roots (Finzi et al., 2015). Thus, the range of the biopore sheath is 0–2 mm from the biopores. This is mainly because the content of nutrients in the 0–2 mm region of the biopore sheath is usually higher than that in the 2–4, 4–8, and 8–12 mm regions (Petzoldt et al., 2020). However, the size of the biopore sheath may be highly variable and related to the biopore size. Coarse roots can effectively move soil particles due to their higher strength and are hence able to influence much larger areas of the surrounding soil than fine roots (Clark et al., 2003). Compared to the bulk soil, the oxygen concentration, water transport, and gas exchange in and around biopores are considerably higher (Colombi et al., 2017; Edwards et al., 1990; Uteau et al., 2013). Pierret et al. (1999) found that the microbial biomass and concentrations of N and P in the biopore sheath soil were higher than those in the bulk soil. Therefore, it takes only a short distance for the water, air, and nutrients provided by biopores to be used by plant roots in the biopore sheath (Stewart et al., 1999).

3 | ROOT-BIOPORE INTERACTIONS

3.1 | Behavior of roots encountering biopores

Several studies found that in well-structured soils, the biopores have little effect on the root system. Xiong, Zhang, Wang, et al. (2022) observed that maize roots did not tend to colonize the biopores in loose soil. Under good aeration conditions, Dexter (1986) found no evidence of preferential root growth toward the biopores. However, plant roots can use biopores, whether artificial or natural, as a pathway of low mechanical resistance to reach the subsoil in unfavorable soil conditions (Colombi et al., 2017; White & Kirkegaard, 2010; Zhou et al., 2021). A concept of “trematotropism” was proposed by Dexter (1986), which described the behavior of plant roots to sense biopores and grow preferentially into them. The presence of biopores can increase the oxygen concentration and air permeability of soil (Colombi et al., 2017). Thus, the oxytropic behavior of roots may lead to growth toward biopores because the higher oxygen content is close to them (Pfeifer et al., 2014; Porterfield, 2002). Furthermore, soil strength has a significant impact on the behavior of roots (Konôpka et al., 2009; Tracy et al., 2012, 2013). The roots usually colonize the biopores and change their growth direction when the soil is compacted (1.6 g cm^{-3}), but the roots only cross the biopores without changing their direction of growth in non-compacted soil (1.2 g cm^{-3}) (Atkinson et al., 2020). Xiong, Zhang, Wang, et al. (2022) also found that roots tended to grow in biopores when the soil penetration resistance was greater than 2 MPa, whereas biopores were not necessary for root growth when they were less than 2 MPa. In a field study, Zhou et al. (2021) observed that the proportion of roots in biopores increased with soil depth due to the impact of soil strength (overburden pressure). Moreover, the behavior of plant roots encountering biopores is highly variable with pore characteristics. Dexter (1986) reported that the proportion of roots entering the biopores decreased with decreasing biopore diameter. Xiong, Zhang, Guo, et al. (2022) found that the number of biopores colonized by roots

in compacted soil increased with increasing pore density. The result verified that the probability of roots encountering biopores increased with biopore density (Dexter, 1986). The tortuosity of the biopores may affect the growth direction of the roots in the pores (Bastardie et al., 2002). Compared to biopores with an angle of 40° from the horizontal, a higher number of roots that grew completely in vertical biopores were reported by Hirth et al. (2005). The roots tended to leave the pores when the pore angle was close to the horizontal plane (Hirth et al., 2005). In conclusion, the behavior of roots encountering biopores can be influenced by multiple factors including soil strength, oxygen stress, and pore characteristics (Colombi et al., 2017; Dexter, 1986; Xiong, Zhang, Wang, et al., 2022).

Apart from remaining in the biopores, Athmann et al. (2019) reported that approximately 20%–25% of the faba bean and barley roots in the biopores can pass through the biopore wall and enter the biopore sheath using in situ endoscopy. When the roots leave the biopores, Petzoldt and Kautz (2021) found that the root length density of spring oilseed rape in the 0–2 mm biopore sheath was larger than that in the 2–4 and 4–8 mm biopore sheaths due to the enrichment of N in the 0–2 mm biopore sheath. Thus, the biopore sheath also has a certain attraction to the plant roots, and this attraction is related to the distance from the biopores.

3.2 | Root-biopore interaction is affected by crop species

The effects of biopores on root growth are different for various plant species (Athmann et al., 2019; Perkons, 2018). For example, Athmann et al. (2013) found that barley and oilseed rape had different ways of exploring biopores, which might be related to their different root architectures. Various behaviors were observed when the roots of maize, wheat, and soybean encountered biopores (Colombi et al., 2017). In a pot experiment, more wheat roots crossed the biopores than maize and soybean roots. The number of colonizing roots in biopores for soybean and maize was greater than that for wheat. Differences in the behavior of roots encountering biopores among crop species may be related to their tolerance to soil strength (Colombi et al., 2017). It has been observed that the tolerance of crops to increased mechanical impedance was on the order of wheat > soybean > maize (Bushamuka & Zobel, 1998; Busscher et al., 2000; Grzesiak et al., 2014). Athmann et al. (2019) visualized root development in the biopores of four crop species. Specifically, many oilseed rape roots grew vertically in the biopores and had good contact with the biopore walls. Most of the cereal roots (e.g., barley and wheat) crossed the biopores horizontally and some of them were in contact with the biopore walls. Almost all the faba bean roots crossed the biopores without contact with the biopore walls. Wahlström et al. (2015) observed that fodder radish tended to grow along the tube (biopore) wall more than wheat. Huang et al. (2020) found that the increase in root diameter of barley was greater than that of rape at high biopore density, which was associated with a higher percentage of barley roots subject to radial swelling than canola roots. Thus, the plasticity of root architecture in

response to soil conditions may have a major influence on root–biopore interactions.

Except for biopores, different crop species also have various responses to the biopore sheath. Petzoldt and Kautz (2021) observed that the utilization of oilseed rape roots to the soil near the biopores was greater than that of faba bean, possibly due to their various rooting behavior. Moreover, Petzoldt et al. (2020) found that the root length density of spring barley was 7.4 times higher than that of faba bean at a distance of 0–2 mm from the biopores. This may be due to the thicker and less morphological plasticity of faba bean roots (Athmann et al., 2019; Li et al., 2014; Materechera et al., 1991; Zhang et al., 2016), resulting in fewer roots in the biopore sheath.

3.3 | Root growth is affected by biopores

Soil compaction usually limits root elongation and slows the development of root systems (Bengough et al., 2011; Correa et al., 2019). Biopores are favorable for plant roots to bypass compacted soil and reach deep soil quickly (Atkinson et al., 2020; Colombi et al., 2017; Pfeifer et al., 2014). Thus, root volume and shoot dry weight increase in compacted soil in the presence of biopores (Pfeifer et al., 2014). Colombi et al. (2017) reported that the detrimental effect of compaction on seedling vigor could be compensated by the effect of biopores in the later growth stage. Han, Kautz, Perkons, Uteau, et al. (2015) found that, although the biopores provide pathways for wheat root growth, the root length inside of biopores was lower than that outside of biopores. For the roots outside of biopores, Petzoldt et al. (2020) found that root length densities of spring barley and faba bean decreased with increasing distance from the biopores, with the crop roots preferring to grow in the biopore sheath. Thus, the biopore sheath also provides a favorable environment for plant root growth. Han, Kautz, Perkons, Uteau, et al. (2015) revealed that the root length of wheat inside biopores was greater in chicory (taproot)–wheat sequences than in tall fescue (fibrous root)–wheat sequences because more biopores were formed by chicory. Moreover, roots with different size classes have different responses to biopore size. The small biopores promote fine root (<0.2 mm) growth, and the large biopores increase the root-length density of thick roots (>0.2 mm) (Han et al., 2016). Furthermore, Kautz (2015) reported that the density of biopores also had an important effect on crop growth. Under 30% water holding capacity, Koch et al. (2021) found that the shoot biomass of wheat increased when the biopore density increased, but not for root biomass. At a given biopore diameter of 0.5 mm, Xiong, Zhang, Guo, et al. (2022) reported that a biopore density of 7000 m⁻² might better enhance maize root growth than biopore densities of 1750 and 28,000 m⁻² in compacted soil. Therefore, biopore characteristics such as biopore diameter and density may have a significant effect on crop performance.

However, the effects of biopores on root growth are not always positive. Some studies have shown that biopores with large diameters do not promote or even limit root growth (Bauke et al., 2017; Stirzaker et al., 1996). Passioura (2002) found that the growth of roots in bio-

pores might be limited when the diameter of biopores was much larger than that of the roots. Except for the biopores, the biopore sheath may also inhibit root growth, possibly due to the dense arrangement of soil particles near the biopores (Schrader et al., 2007). For example, once the roots enter the biopores, they may not be able to re-enter the bulk soil (Stirzaker et al., 1996). Jakobsen and Dexter (1988) reported that the crop yield decreased as the biopore diameter was larger than 0.4 mm in a computer simulation study.

We further summarized data regarding the effects of biopores on shoot or root growth from nine studies (Table 2). The summary suggests that the biopores promoted shoot or root growth in five of nine studies but did not increase crop growth in the other four studies. These findings show that the biopores with diameters larger than or equal to 2 mm had no positive effect on plant growth, and the biopores with diameters less than 2 mm promoted shoot or root growth. Thus, our review demonstrates that the effect of biopores on plant growth might be related to biopore diameter, which determines the contact between the soil and root. Various studies have found that poor root–soil contact in large biopores might restrict plant growth (Bauke et al., 2017; Passioura, 2002; Stirzaker et al., 1996). The presence of air gaps between roots and soil also limits the ability of roots to absorb water and nutrients due to poor root–soil contact (Passioura, 2002; Veen et al., 1992). Bauke et al. (2017) found that biopores 6 mm in diameter did not promote the absorption of nutrients (overall P), possibly due to a lack of contact between the roots and soil, although the biopores affected the distribution of ³³P in the roots. However, this explanation still needs to be confirmed by quantifying root–soil contact in biopores. Furthermore, root growth in large biopores may release some inhibitory signals to influence stomatal conductance and leaf appearance, thus affecting plant growth (Passioura, 2002).

3.4 | Root absorbing nutrients and water are affected by biopores

Deep rooting is a crucial trait for plants to access water resources in the subsoil (Lynch, 2013). Rasmussen et al. (2020) found that chicory roots can grow to a depth of 3 m and can absorb large amounts of water from the soil below 2 m, regardless of whether there is sufficient water. Yunusa and Newton (2003) reported that plants with strong penetrating ability could improve the soil pore structure and facilitate rapid access to soil water for subsequent crops. The presence of biopores promotes rooting, which is beneficial to the uptake of water by plants (Kautz, 2015). In arid areas, biopores can attract roots to deeper soil to access water (McKenzie et al., 2009). Using a three-dimensional soil–root model (R-SWMS), Landl et al. (2019) found that the water uptake by roots increased during the drought period in the presence of biopores, especially in compacted and low-conductivity soils. Moreover, Aravena et al. (2011) revealed that root-induced soil compaction enhanced the connectivity between the aggregates, which improved the effective hydraulic conductivity of the rhizosphere and facilitated water uptake by roots. The water content of the soil around the roots is usually higher than that of the bulk soil during drought due

TABLE 2 Biopore effects on shoot or root growth by location, soil texture, soil bulk density, plant species, biopore diameter, and biopore density.

Location	Soil texture	Bulk density (g cm ⁻³)	Plant species	Biopore diameter (mm)	Biopore density (cm ⁻²)	Positive effects	Reference
Alberta, Canada	Clay loam	1.42	Wheat	0.5–1	0.03–0.45	Yes	Volkmar (1996)
China	Clay loam	1.6	Maize	0.5, 1	0.18	Yes	Xiong, Zhang, Guo, et al. (2022)
Germany		1.64	Barley	1	0.57	Yes	Pfeifer et al. (2014)
NSW, Australia	Sandy clay loam	1.5	Tall wheatgrass	1	0.15	Yes	Haling et al. (2011)
Zurich, Switzerland	Silt loam	1.6	Soybean, wheat, maize	1.25	0.27	Yes	Colombi et al. (2017)
China	Clay loam	1.6	Maize	2	0.19	No	Xiong, Zhang, Wang, et al. (2022)
Australia	Sandy loam	1.77	Barley	3.2	0.41	No	Stirzaker et al. (1996)
Australia		1.78	Barley	3.2	0.41	No	Passioura (2002)
Germany	Silty loam	1.4	Wheat	6	0.04	No	Bauke et al. (2017)

Note: Positive effects mean that the presence of biopores promotes shoot or root growth.

to the influence of root exudates (Carminati et al., 2010; York et al., 2016). Thus, the biopore sheath may have higher water retention after root death, and consequently influence water uptake.

Over 60% of soil nutrients are located in the subsoil, with the presence of biopores facilitating root elongation and promoting nutrient uptake in the subsoil (Kautz, 2015; Kautz et al., 2013). Han et al. (2017) found that an increase in small biopores enhanced the proportion of fine roots in deep soil, which improved the absorption of nitrogen and phosphorus by canola. Using radioactive ³³P isotopes, Han et al. (2022) also reported that the larger proportion of total P uptake by plants from deeper soil layers might be related to more fine roots growing in the subsoil. Therefore, it is a good strategy to obtain nutrients from subsoil through biopores, especially in compacted subsoils (Kautz et al., 2013). Moreover, the decomposition of roots was a source of nutrients in the biopore (root channels) sheath (Smucker, 1993). Athmann et al. (2017) reported that biopore sheaths contained many nutrients, such as N, S, and P, and their contents were two times higher than that of bulk soil. Through contact between the root and the biopore wall, the accessibility of nutrients to roots can be increased (Kautz, 2015). The entry of fine roots into the biopore sheath is also significant for nutrient absorption in subsoil (Petzoldt et al., 2020). Petzoldt and Kautz (2021) found that oilseed rape roots could grow toward the biopore sheath due to N enrichment in the biopore sheath. Athmann et al. (2016) reported that wheat could absorb 32% of its nitrogen from the biopore sheath of the subsoil. Thus, the biopore sheath plays an important role in nutrient absorption.

However, many studies have found that the clumping of roots in large-sized biopores may limit the absorption of nutrients and water (Cresswell & Kirkegaard, 1995; White & Kirkegaard, 2010). When the roots grow in biopores with large diameters, even though the soil water is sufficient, the ability of roots to access water may also be restricted due to poor root–soil contact (Passioura, 2002). Using X-ray computed tomography (CT), Xiong, Zhang, Wang, et al. (2022) observed that biopores with a diameter of 2 mm had little impact on the nutrient uptake

of maize from the subsoil because the maize root diameter was much lower than the biopore diameter, resulting in poor root–biopore wall contact. Xiong, Zhang, Guo, et al. (2022) also found that 0.5 and 1 mm biopores promoted ¹⁵N uptake of maize shoots compared with 2 mm biopores. Therefore, the establishment of good contact between the root and biopore wall to make full use of the advantages of biopores is very important for promoting crop growth. Furthermore, quantifying the contact between the root and biopore wall is urgently needed, which helps to test this explanation. However, it is still unclear how biopore (sheath) characteristics affect the nutrient uptake of plant roots (Petzoldt et al., 2020).

4 | METHODS FOR QUANTIFYING ROOT-BIOPORE INTERACTION

In the past, it was impossible to conduct in situ investigation of root behavior in the soil because the soil matrix was opaque (Heeraman et al., 1997). Therefore, researchers have to use destructive methods to study the relationship between roots and soil (Young et al., 2001). However, these destructive techniques can hardly reveal the spatial relationship between roots and biopores in situ. Kautz and Köpke (2010) proposed an in situ endoscopy technique to visualize the root behaviors in biopores. Although this method can directly visualize the growth of plant roots in biopores, it is difficult to quantify the interactions between roots and biopores. A study by Tracy et al. (2010) found that X-ray CT could be used for the study of soil–plant interactions, such as noninvasive three-dimensional visualization and measurements. Other methods, such as neutron radiography, can also be used to study root–soil interactions, but this method is often used to observe water movement between roots and soil because of its sensitivity to water (Carminati et al., 2010). Therefore, X-ray CT will be a superior method to study root–biopore interactions in the future, as it is more effective in imaging soil structure and plant roots.

4.1 | Destructive observation

In the past, many researchers quantified the interactions between roots and soil using destructive techniques (Bodner et al., 2014; Chen & Weil, 2011; Gaiser et al., 2012; Kirkegaard & Lilley, 2007; Perkons et al., 2014). Kooistra et al. (1992) analyzed the degree of contact between roots and soil when roots grew in biopores using a thin-section method. White and Kirkegaard (2010) used a microscope to observe wheat roots growing in biopores in broken soil cores. Petzoldt et al. (2020) sampled the biopore sheath of different distances from the biopores using the biopore sheath scraping device. Although destructive methods destroy the soil structure and require time, they can provide useful information when combined with other technologies.

4.2 | In situ endoscopy

To perform in situ observations of root growth in biopores, Kautz and Köpke (2010) proposed an in situ endoscopy method. The principle of endoscopy is to use a flexible videoscope to assess the behavior of roots in biopores. Specifically, it can be used to observe whether the vertical roots contact the pore wall or whether the horizontal roots re-enter the bulk soil. Using the endoscopy method, the characteristics of the biopore wall and the distribution of roots in biopores were clearly visible (Athmann et al., 2013). Similar to an in situ detection method, endoscopy can visualize individual large biopores (≥ 6 mm) (Kautz & Köpke, 2010), but it is difficult to apply if the pore network is complex.

4.3 | Neutron radiography

Neutron radiography is one of the few noninvasive methods that can be used for in situ imaging of living plant roots and water distributions (Carminati et al., 2010; Menon et al., 2007; Oswald et al., 2008). Although neutron radiography mainly focuses on water dynamics in soil, it is still helpful for the study of root–biopore interactions. For example, the amount of water absorbed by the roots in biopores and the effect of different sizes of biopores on root water absorption can be quantified by neutron radiography.

4.4 | X-ray CT

In recent years, the use of X-ray CT has made it feasible to study the interactions between roots and biopores in situ. Many researchers have reported the application of X-ray CT in quantifying root–soil interactions (Koebernick et al., 2018; Kuka et al., 2013; Mairhofer et al., 2012; Tracy et al., 2012; Xiong et al., 2020). For instance, using X-ray CT, Carminati et al. (2009) observed the formation of gaps between roots and soil during soil drying. However, there are few studies related to directly quantifying the behavior of roots encountering biopores using X-ray CT (Atkinson et al., 2020; Colombi et al., 2017; Pfeifer et al.,

2014). Quantification of root–biopore interactions, especially the contact area between the root and soil in biopores, is still very difficult due to the limitation of the X-ray CT resolution. The gaps between root and soil were also difficult to extract accurately from images. Therefore, it is a challenge to quantify the contact area between roots and pore walls. In future studies, the CT machines that can scan large columns with higher spatial resolution are needed.

5 | KNOWLEDGE GAPS AND FUTURE WORKS

Biopores have received more attention due to their various functions, such as enhancing the uptake of nutrients and water by roots (Putten et al., 2004). Many studies have reported the biopore effects on plant growth, but the results are inconsistent (Bauke et al., 2017; Colombi et al., 2017; Pfeifer et al., 2014; Stirzaker et al., 1996). In the field, the size, shape, and direction of biopores are highly variable due to the various root architectures of different crop species (Pagenkemper et al., 2013). Therefore, these characteristics of biopores may play a key role in affecting crop growth. To date, the effects of biopore characteristics on plant growth are still not clear (Jin et al., 2017). Determination of the appropriate characteristics of biopores to promote root development is urgently needed. Once suitable biopore characteristics are determined, these beneficial biopores can be produced in the field by bio-tillage using cover crops, which is important for sustainable agriculture (Zhang & Peng, 2021).

In the process of roots seeking biopores, some concepts have been put forward, such as “trematropism” and “oxytropism” (Dexter, 1986; Porterfield & Musgrave, 1998). To date, a possible explanation for these behaviors is that the roots are able to sense the spaces with the least mechanical resistance or a higher oxygen concentration, which can be provided by biopores in unfavorable soil conditions (Colombi et al., 2017; Porterfield, 2002). Except for the mechanical resistance and oxygen content, other factors may also affect the exploration of biopores by roots. For example, the lack of pores in compacted soil leads to a low air diffusion rate and consequently the accumulation of ethylene around the root tips, which limits plant growth (Pandey et al., 2021). We speculate that the plant roots can sense soil biopores by detecting the ethylene concentration distribution, which should be lower in the space close to the biopores due to the increased diffusion rate of the biopores. This hypothesis is expected to be verified in future studies. In general, the mechanism for the behavior of plant root growth toward biopores is still unclear and requires more investigation.

To date, many studies have focused on the effect of biopores on root growth and the behaviors of roots in biopores (Atkinson et al., 2020; Colombi et al., 2017; Pfeifer et al., 2014). Plant roots not only create biopores, but also affect the soil structure and nutrient contents in the biopore sheath (Lucas et al., 2019a, 2019b). The biopore sheath soil also plays a vital role in crop growth due to its high nutrient content and water retention (Pierret et al., 1999). In future research, we should combine the biopores and the biopore sheath as a whole to investigate their effects on crop growth.

6 | CONCLUSIONS

Plant rooting can create channels in the soil and improve soil structure, in which biopores are formed when the roots decompose. Root-derived biopores are usually cylindrical, continuous, and stable, with good conductivity of air and water, which are different from other types of soil pores. We proposed the concept of a “biopore sheath” to define the soil affected by roots and resulting biopores. Roots may cross or colonize biopores, depending on soil strength, oxygen stress, biopore characteristics, and plant species. Root-derived biopores can provide minimal resistance pathways for roots to reach the subsoil, which may promote the uptake of nutrients and water. When the roots leave the biopores and re-enter the soil, they tend to grow to the biopore sheath due to the enrichment of nutrients. However, the effect of biopores on plant growth is closely related to the characteristics of biopores (e.g., biopore diameter). From the collected literature, we found that smaller diameter biopores (<2 mm) promote plant growth, while larger diameter biopores (≥2 mm) do not. Good root–soil contact in biopores may be an essential condition for biopores to improve crop growth. In recent years, X-ray CT technology has achieved remarkable success in the quantification of the interactions between roots and biopores. However, precisely quantifying the contact between roots and soil in biopores is still a challenge due to the limitation of CT image resolution. Furthermore, the biopore characteristics suitable for crop root growth and nutrient uptake and the mechanisms of root growth toward biopores are not entirely clear. Thus, these gaps should be addressed to clarify the root and root-derived biopore interactions.

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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